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Genetic Assessment of Walleye (*Sander vitreus*) Restoration Efforts and Options in Nipigon Bay and Black Bay, Lake Superior

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ABSTRACT. Walleye (*Sander vitreus*) stocks in Nipigon Bay and Black Bay historically numbered as the largest stocks in Lake Superior, but collapsed in the 1960s due to overfishing, habitat loss, and other pressures. We used microsatellite DNA analyses to assess the success and relative contributions of past rehabilitation stocking to walleye in Nipigon Bay, and to investigate the relationship between historical and contemporary populations in Black Bay. Based on the genetic data, juvenile stocking and adult transfers from four source populations into Nipigon Bay differed substantially in their contributions to the re-established population. The genetic data also indicated that natural reproduction was occurring and identified survivors from the former Nipigon Bay population. Similar genetic analysis of scale samples from the historical Black Bay fishery and present-day walleye from a major tributary (Black Sturgeon River) showed that the historical and contemporary samples comprise one genetic stock, which is significantly different from neighboring native and introduced populations. These findings suggest that walleye restoration efforts in Lake Superior are working, and highlight the utility of and options for adaptive management approaches for restoring extirpated populations.

INDEX WORDS: Restoration, walleye, *Sander vitreus*, genetics, microsatellite DNA.

INTRODUCTION

Fisheries management in the Great Lakes has undergone a significant transformation over the past century. Although management practices of 100 and even 50 years ago focused on maximizing exploitation and harvest efficiencies, the past several decades have seen a significant shift toward true sustainable management. With increasing challenges and lessons learned, management practices have evolved to include proactive conservation of unstable or at-risk fisheries resources and restoration of extirpated populations. Notable achievements include the ongoing recovery of lake trout in the Great Lakes (Selgeby *et al.* 1995, and refer-

ences therein) and current rehabilitation efforts for coaster brook trout in Lake Superior (Huckins *et al.* 2007, Schreiner *et al.* 2007, Wilson *et al.* 2007).

In Lake Superior, Nipigon Bay and Black Bay (Fig. 1) historically supported the largest fisheries of walleye, *Sander vitreus* (Schneider and Leach 1977). Recorded harvests date back to the mid-nineteenth century (Wilson 1991), and these two populations provided 90% of the commercial walleye harvest in Lake Superior from 1948 to 1974 (Schneider and Leach 1977). Although the Black Bay fishery was larger by far, both populations suffered catastrophic collapses within a single generation (Schneider and Leach 1977).

Walleye numbers in Nipigon Bay collapsed in 1965, most probably due to a combination of stressors (Wilson 1991). The loss of the population was attributed to pollution and habitat degradation (Ryder

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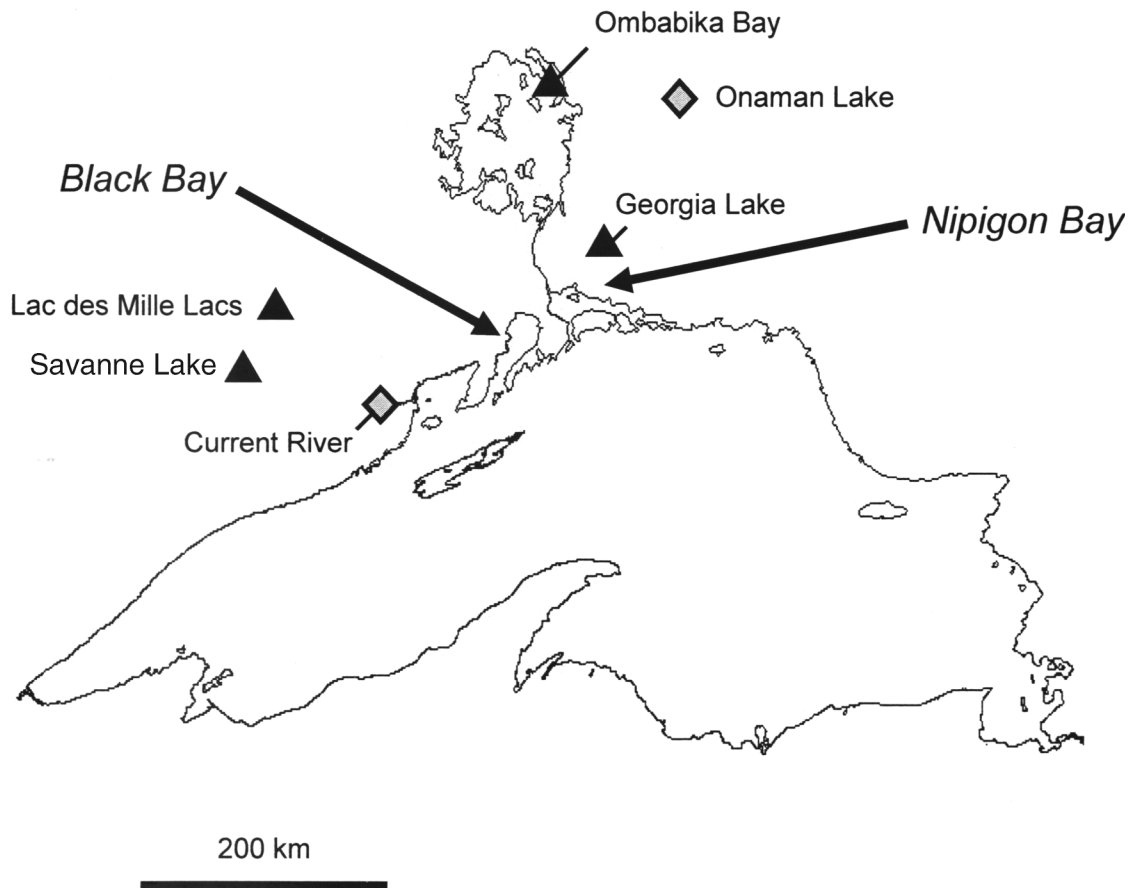


FIG. 1. Map of Lake Superior, showing study locations and sources for stocking or adult transfer. Adult transfer source populations are indicated by triangles [Georgia Lake (49°19' N; 87°53' W); Lac des Mille Lacs (48°51' N; 90°30' W); Ombabika Bay, Lake Nipigon (50°16' N; 88°22' W); and Savanne Lake (48°50' N; 90°06' W)]. Black diamonds indicate previous stocking sources listed in Table 1 [Current River (48°27' N; 89°11' W); Onaman Lake (50°01' N; 87°26)].

1968), although heavy commercial fishing may have played a role as well (MacCallum and Selgeby 1987). Walleye in Black Bay declined catastrophically in the 1960s (Hoff 2003), dropping from a peak harvest of 170 metric tonnes in 1966 to zero by 1972 (MacCallum and Selgeby 1987). The collapse in Black Bay was attributed to overharvest (Colby and Nepszy 1981), although other factors may have been involved (MacCallum and Selgeby 1987). Restoration efforts were initiated soon after the dual fishery collapses and are still ongoing, with both sites identified as priority sites for remediation and restoration (Colby *et al.* 1991, 1994; Hoff 2003).

Nipigon Bay

Rehabilitative stocking of early life stages of walleye into Nipigon Bay from 1978 to 1986 (Table

1) had little apparent success (Wilson 1991). Population assessment throughout the 1980s failed to detect significant numbers of walleye (Wilson 1991). Subsequent large-scale efforts involved collection and transfer of reproductively mature walleye into Nipigon Bay from Savanne Lake (3,600 adults between 1987 and 1989), and 12,000 adults from three sources (Ombabika Bay, Lake Nipigon, Georgia Lake, and Lac des Mille Lacs; Fig. 1) between 1990 and 1992 (Table 1). For each adult transfer event, reproductively mature adults were collected from spawning aggregations in the source habitats. Genetic analysis of the three latter source populations showed that they were recognizably different (Ihssen *et al.* 1993), thereby allowing for subsequent assessment of relative stocking success among strains.

TABLE 1. History of rehabilitative stocking efforts into Nipigon Bay, showing stocking years, source populations, life stage stocked, and numbers (data from Wilson 1991 and OMNR district records).

Year	Source	Life stage	Number stocked
1978	Current River	eggs	500,000
1979	Current River	eggs	1,170,000
1980	Current River	eggs	1,364,000
	Current River	fingerlings	20,800
1981	Onaman Lake	eggs	500,000
	Onaman Lake	fingerlings	3,700
1982	Onaman Lake	eggs	1,461,000
	Onaman Lake	fingerlings	11,755
1983	Onaman Lake	eggs	2,450,000
	Onaman Lake	adults	347
1984*	Current River	eggs	350,000
	Onaman Lake	eyed eggs	200,000
	Lake Nipigon	fry	757,000
1985	Lake Nipigon	fry	4,000,000
	Lake Nipigon	fingerlings	1,800
1986	Savanne Lake	fry	425,000
	Savanne Lake	adults	556
1987	Savanne Lake	adults	421
1988	Savanne Lake	adults	932
1989	Savanne Lake	adults	777
1990	Lac des Mille Lacs	adults	3,500
1991	Georgia Lake / Lac des Mille Lacs / Lake Nipigon	adults	4,400
1992	Georgia Lake / Lac des Mille Lacs / Lake Nipigon	adults	4,200

* numbers reflect total (pooled) numbers of fish from multiple sources stocked at each life stage.

Black Bay

Current restoration efforts are focused on identifying suitable spawning habitat and source populations for stocking. Although the historical population was thought to consist of shoal spawners, the Black Sturgeon River may have been the actual spawning habitat (Colby and Foster 2001). A logging dam built 16 km upstream from the river mouth in the 1950s (Fig. 2) blocked access to the upper reaches of the river, and preceded the population collapse by only 5 years (Furlong *et al.* 2006). Stocking efforts in recent years have utilized diverse sources (Colby and Foster 2001), but the per-

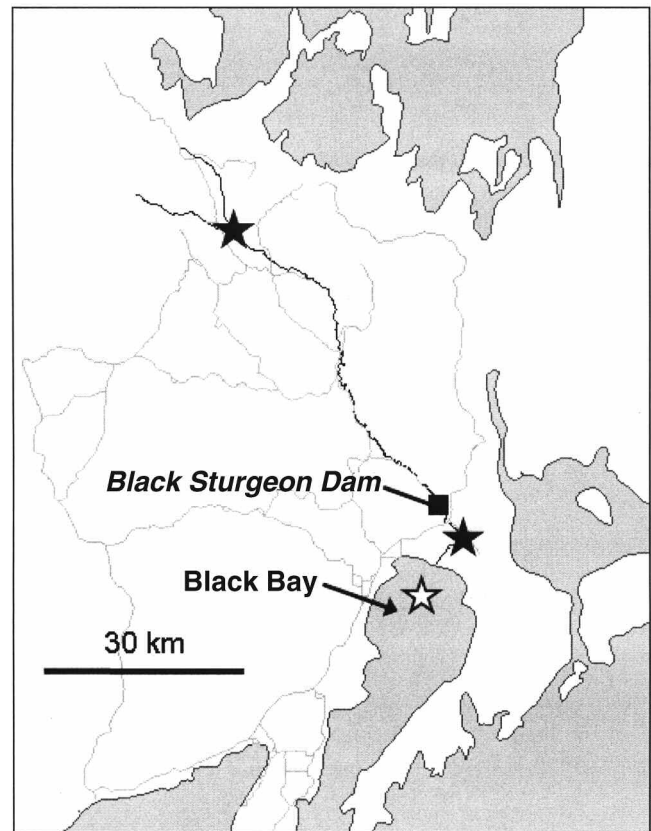


FIG. 2. Black Bay and the Black Sturgeon River, showing sampling locations for genetic analysis. The dam on the lower Black Sturgeon River is indicated by the filled black box. Black stars indicate sampling locations in the Black Sturgeon River from 2002 and 2003; the white star represents historical samples from the 1966 commercial walleye fishery in Black Bay.

sistence of walleye in the Black Sturgeon River raises the possibility that the historical Black Bay population may not have been completely extirpated (Furlong *et al.* 2006).

Widely recognized, clear guidelines have been laid out for fisheries restoration programs (Krueger *et al.* 1981, Meffe 1995). Where possible, genetic relatives of the depressed or extirpated population should be used in rehabilitation or restoration efforts (Krueger *et al.* 1981). In some cases, this is not possible without data from the lost population, and managers must choose between inferred shared ancestry and allopatric sources with suitable ecological traits (Meffe 1995). Recent approaches to restoration biology have suggested setting out explicit contrasting hypotheses, such as testing eco-

logical suitability versus shared ancestry in restoration programs (Moritz 1999, Jones 2003). Each of these approaches requires a genetic component to monitoring or assessment efforts, in order to evaluate the overall and relative success of rehabilitation programs. As well as providing information on establishment of introduced strains (e.g., Murphy *et al.* 1983), genetic tracking is a powerful assessment tool for evaluating the relative performance of different strains (Eldridge *et al.* 2002).

We employed neutral genetic markers (microsatellite DNA) to assess the success of past rehabilitation stocking and support current efforts in both Nipigon Bay and Black Bay. The relative contributions and fitness of the four source populations to the recovering walleye population in Nipigon Bay were assessed using both population- and individual-based analyses. Similar approaches were used to resolve the relationship between the historical walleye population in Black Bay and the contemporary walleye population in its major tributary, the Black Sturgeon River, above and below a man-made barrier dam. The results from both studies suggest a role for both ecological and genetic criteria in decision making for adaptive management approaches to population restoration.

METHODS

Tissue samples from the four source populations used for adult transfers into Nipigon Bay were provided by the OMNR Upper Great Lakes Management Unit (Table 1). Samples were provided as archival fin ray samples previously used for aging, as well as ethanol-preserved finclips. Walleye captured in Nipigon Bay between 2000 and 2003 were nonlethally sampled using finclips for tissue sources for genetic analysis. To resolve the relationship between walleye in the Black Sturgeon River and the historical Black Bay population, archival scale samples from the 1966 commercial fishery ($n = 80$) were provided by the OMNR Upper Great Lakes Management Unit. Tissue samples were also obtained from 85 walleye sampled in 2003 and 2004 above and below the dam on the Black Sturgeon River using nonlethal finclips (Fig. 2).

Genomic DNA was extracted from finclips and historical scale samples using a crude lysis extraction method. Tissues (approximately 10 mg fin tissue, or 2–6 scales) were lysed in 96-well plates using 250 μ L lysis buffer (50 mM Tris pH 8, 100 mM NaCl, 1 mM ethylenediamide tetraacetic acid [EDTA], 1% sodium dodecyl sulphate weight per

volume, and 100 μ g proteinase K) per well, and incubated for 16 hours at 37°C. Genomic DNA and degraded proteins were precipitated using 500 μ L of 80% isopropanol per well and centrifugation at 2,000 gravities (19,600 m/s^2) for 30 minutes. Supernatant was removed and the remaining pellets were rinsed with 1 mL of 70% ethanol, followed by re-centrifugation. DNA pellets were air-dried at room temperature for 20 minutes, then resuspended in 50–200 μ L of TE buffer (10 mM Tris pH 8, 1 mM EDTA). Extraction yields and quality were tested using horizontal electrophoresis in 0.8% agarose gels alongside a Fermentas™ molecular mass ladder.

Multilocus microsatellite genotypes were obtained from each sample for five microsatellite loci. The five fluorescently-labeled loci (*Svi2*, *Svi4*, *Svi6*, *Svi7*, and *Svi14*; Borer *et al.* 1999) were co-amplified in a single 10 μ L multiplex polymerase chain reaction (PCR) containing sterile deionized water, 1X manufacturer buffer, 1.5 mM $MgCl_2$, 0.2–0.6 pmol of each primer pair, 200 μ M dNTPs, 0.25 units *Taq* polymerase, and approximately 8 ng of genomic DNA. The PCR cycling program consisted of 2 minutes initial denaturation at 94°C, followed by 30 cycles of 30 seconds at 94°C, 1 minute at 60°C, 1 minute at 72°C, and a final extension time of 40 minutes at 60°C. Amplified products were electrophoresed in a 5% polyacrylamide gel on an ABI Prism 377 DNA Sequencer and sized using ROX 350 size standards (Applied Biosystems Inc.). Genotypes at each locus were scored using Genotyper 2.0 (Applied Biosystems Inc.) and confirmed by manual proofreading.

Data Analysis

Nipigon Bay

Genetic relationships among populations were estimated using D_A (Nei *et al.* 1983) based on population allele frequency estimates. Pairwise genetic distances were calculated and used to construct a neighbour-joining dendrogram using the program Populations (Langella 2002); support for the tree topology was estimated using 1,000 bootstrap replicates. The resulting dendrogram was plotted using Treeview (Page 1996).

Ancestry of walleye collected from Nipigon Bay in 2000–2003, as well as those from the four source populations, was determined using individual assignment tests (Cornuet *et al.* 1999, Pritchard *et al.* 2000). Genetic ancestry of individuals was assessed

using a partial Bayesian approach (Rannala and Mountain 1997) as implemented in GeneClass 2.0 (Piry *et al.* 2004). Individuals were considered as reliably assigned to a single source population if the probability of their genotype occurring by chance in that population, based on baseline allele frequency data, was significantly greater than that for the next most probable source (*i.e.*, if the log-likelihood ratio ≥ 2). These assignments were tested by complementary exclusion analyses, using simulated population sizes of 10,000 fish for each source population (Cornuet *et al.* 1999) to confirm exclusion of individual genotypes from non-source populations at a rejection level of $p = 0.01$. These combined analyses allowed for several possible outcomes, allowing individuals to be assigned to either a single source population (sole ancestry), to two or more populations due to mixed ancestry, or to be excluded from all known source populations. Assignment interpretations were independently assessed using Structure 2.1 (Pritchard *et al.* 2000), which also allowed estimation of mixed-ancestry contributions for fish captured in Nipigon Bay. Independent analyses were run with and without using population identifiers for fish sampled from the source populations for comparison; all fish sampled from Nipigon Bay were considered as unknowns. The program Structure was also used to estimate the number of genetic groups (K) present in the Nipigon Bay and sources dataset, using the complementary hypotheses of the recovering population being exclusively derived from introductions from the four source populations (*i.e.*, all sampled walleye in Nipigon Bay originating or descended from the introduction sources), versus the presence of non-introduced (novel) genotypes that could be excluded from all source populations and their possible combinations, suggesting the presence of remnant or immigrant walleye.

Black Bay

The relationship between the historical Black Bay population and walleye presently found in the Black Sturgeon River was similarly assessed using population- and individual-based approaches, with walleye from the recovering Nipigon Bay population included as an outgroup for comparison. Allele frequencies within each sample set (historical samples, and above- and below-barrier subsets from the Black Sturgeon River) were pooled to enable population-level comparisons using the D_A genetic distance estimator (Nei *et al.* 1983). The inter-

population comparison was complemented with individual assignment and exclusion tests using Bayesian resampling methods (Cornuet *et al.* 1999, Pritchard *et al.* 2000). The number of genetic groupings (K) within Black Bay and its major tributary were assessed using complementary hypotheses of (i) panmixia (one group), (ii) separate bay versus river populations (two groups), and (iii) a separate bay population plus substructure above and below the river dam (three groups). Unknowns from Nipigon Bay were again included as an outgroup for comparison. Analytical runs in Structure 2.1 relied on 50,000 sampling cycles prior to data collection, followed by 100,000 resampling steps, using 10 replicate trials for each estimation of K .

RESULTS

Nipigon River Walleye Restoration

Genotyping of the four adult source populations showed that all five microsatellite loci were highly polymorphic, and variable among the source populations (Table 2). Of the four populations, Savanne Lake showed the lowest diversity and Lake Nipigon the greatest, possibly reflecting differing native habitat conditions and effective population sizes. All four populations were clearly distinct based on pairwise genetic distance estimates (Fig. 3), corroborating earlier allozyme data by Ihssen *et al.* (1993). These differences were reflected in the individual assignment tests, with high self-assignment scores among the source populations (Table 3). Genotypes of walleye from Lac des Mille Lacs and Savanne Lake resulted in 100% correct assignment to their respective sources despite the limited divergence between the two lakes (Fig. 3), due to the lower genetic diversity and limited number of possible genotypes in these populations (Tables 2 and 3). Conversely, some walleye from Georgia Lake and Ombabika Bay were reciprocally mis-assigned to the other population (Table 3). The mean correct assignment to source for the donor populations using GeneClass2 was 95.5% (Table 3), giving an estimate of the power for assignment of individuals from Nipigon Bay to potential sources.

Individual assignment tests of walleye captured in Nipigon Bay showed highly unequal contributions of the different source populations (Table 3). Because a maximum of two generations (16 years) had passed since all four stocks were introduced, 25% was set as a minimum threshold to infer ancestry from any of the introduced populations. Ombabika Bay in Lake Nipigon appeared to make the

TABLE 2. Allele frequencies and sample sizes (N) for five microsatellite loci for walleye sampled from the four source populations and recent captures in Nipigon Bay.

	Georgia	Mille Lacs	Ombabika	Savanne	Nipigon Bay		Georgia	Mille Lacs	Ombabika	Savanne	Nipigon Bay
<i>N</i>	54	64	50	47	57	<i>N</i>	54	64	50	47	57
Svi2						Svi7					
192	0.08	0.01	0.05	—	0.32	161	—	0.01	—	—	—
194	0.32	0.65	0.52	0.47	0.22	163	0.39	—	0.17	—	0.23
196	—	0.34	0.19	0.53	0.03	165	0.01	—	0.25	—	0.05
198	0.20	—	0.15	—	0.12	167	0.01	0.03	0.24	0.01	0.36
200	0.37	—	0.07	—	0.09	169	0.06	0.01	—	0.15	0.10
204	0.02	—	—	—	0.02	171	0.42	0.03	0.20	—	0.23
206	—	—	0.02	—	0.18	173	0.11	0.92	0.14	—	0.04
220	—	—	—	—	0.03	175	—	—	—	0.84	—
Svi4						Svi14					
99	—	—	—	—	0.01	160	—	—	—	—	0.01
103	—	—	0.03	—	0.05	170	—	0.01	—	—	—
105	0.01	0.02	—	—	—	174	—	0.01	—	—	0.01
107	0.58	0.58	0.55	0.59	0.35	176	—	—	0.04	—	0.01
109	0.16	0.16	0.07	0.37	0.05	178	—	—	0.02	—	—
111	0.14	0.23	0.10	0.04	0.16	180	—	0.03	0.06	—	0.01
113	0.07	—	0.25	—	0.36	182	0.31	0.15	0.12	0.25	0.11
115	—	0.01	—	—	0.01	184	0.11	0.47	0.13	0.71	0.18
117	0.04	—	—	—	0.01	186	—	0.05	0.02	—	0.09
Svi6						188	0.24	0.10	0.25	0.04	0.42
116	—	—	—	0.26	—	190	0.17	0.16	0.02	—	0.02
122	0.01	—	—	—	—	192	0.09	0.02	0.12	—	0.06
124	0.61	0.94	0.24	0.74	0.55	194	0.06	—	0.10	—	0.07
126	0.01	—	—	—	0.01	196	—	—	0.06	—	—
128	0.13	—	0.06	—	0.05	198	0.01	—	0.02	—	—
130	0.03	—	0.06	—	0.03	200	0.01	—	0.06	—	0.02
132	0.06	0.06	0.07	—	0.02						
134	0.04	0.01	0.12	—	0.04						
136	0.05	—	0.06	—	0.05						
138	—	—	0.05	—	—						
144	—	—	0.01	—	0.01						
146	—	—	0.01	—	0.17						
148	0.01	—	0.15	—	0.03						
150	—	—	0.14	—	0.02						
152	0.06	—	0.02	—	0.02						

greatest contribution, with 25 of the captured fish showing significant assignment to Ombabika Bay versus other potential sources. Georgia Lake had the second greatest contribution, with five fish showing first-generation ancestry. One captured fish had significant assignment (100%) to Lac des Mille Lacs (Table 3, Fig. 4). The estimated age of this fish was 14 years (data not shown), suggesting that it might be a surviving adult from the stocking transfers. No individuals were detected as having

significant ancestry from Savanne Lake, suggesting that walleye stocked from this source were no longer present.

Individual-based assignment and exclusion probabilities revealed the presence of a fifth genetic group, comprised of individuals that were significantly excluded from any of the four source populations (Table 3, Fig. 4). These fish made up a substantial portion of the Nipigon Bay population, and appear to be successfully reproducing (Fig. 4).

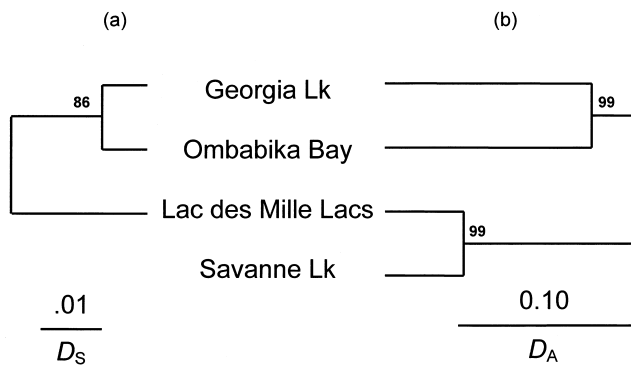


FIG. 3. Neighbor-joining dendrograms of genetic distances among source populations based on (a) 31 allozyme loci (Nei's D_S ; data from Ihssen *et al.* 1993) and (b) five microsatellite loci used in this study (D_A ; Nei *et al.* 1983). Numbers on branch nodes represent percent bootstrap support based on 1,000 resampling events.

In the absence of historical data, it could not be determined whether these fish represented remnants of the original population, stocking from another source, or dispersers from unknown wild populations.

Reproductive contributions of the source populations were similar to their relative stocking success, with many of the mixed-ancestry walleye showing ancestry from both the Ombabika Bay and Georgia Lake populations as well as interbreeding between both groups and the unexpected fifth group (Fig. 4). There was no evidence of assortative mating among the introduced stocks: younger walleye were either descended from Ombabika Bay fish, Ombabika Bay and Georgia Lake mixed-parentage, or matings between Ombabika Bay walleye and the local unknown-origin fish (Fig. 4). No evidence of reproduction by walleye from Lac des Mille Lacs or Savanne Lake was detected. Although it is likely

that differences in local fitness may exist among the source populations, the underlying mechanisms could not be addressed in this study.

Black Bay Walleye Restoration

Results from both the individual- and population-based analyses showed that walleye in the Black Sturgeon River are very closely related to the historical Black Bay population (Table 4, Figs. 5 and 6). Only four of the five microsatellite loci could be used, because amplification of locus *Svi7* was problematic for the historical population. Nonetheless, running Structure (Pritchard *et al.* 2000) to test for one to four genetic groups (panmixia versus all sample sets as distinct genetic groups) resulted in maximum resolution for $K = 2$ clusters [$\ln(\text{PID}) = -1228.5$].

Individual assignment tests using both GeneClass2 and Structure showed that individuals from the historical Black Bay and contemporary Black Sturgeon River sample sets could not be separated, versus all fish from the Nipigon Bay out-group which were consistently excluded from the greater Black Bay dataset (Fig. 5). These results indicate that the Black Sturgeon River and the historical Black Bay population shared a common gene pool, suggesting that the population may not be extirpated, as previously thought. Increasing the number of potential clusters in Structure did not increase resolution within or separation between the Black Bay or Black Sturgeon River sample sets (data not shown). Similarly, some individuals within the historical population and the Black Sturgeon River assigned to the Nipigon Bay group at all estimated values of K , suggesting that their assignment to the Nipigon Bay unknowns reflects shared ancestry rather than a statistical artifact.

Comparison of population-level characteristics corroborated the results of the individual assign-

TABLE 3. Inferred ancestry of individual walleye from source populations and Nipigon Bay based on individual assignment tests using GeneClass2 (Piry *et al.* 2004), showing assessed populations (rows) and inferred genetic ancestry (columns). Table values show ancestry for each category by absolute numbers of fish, with percent contributions in brackets.

	Georgia	Mille Lacs	Ombabika	Savanne	mixed	other
Georgia	51 (94%)	0	3 (6%)	0	—	—
Mille Lacs	0	64 (100%)	0	0	—	—
Ombabika	6 (12%)	0	44 (88%)	0	—	—
Savanne	0	0	0	47(100%)	—	—
Nipigon Bay	5 (9%)	1 (2%)	25 (45%)	0	9 (16%)	17 (30%)

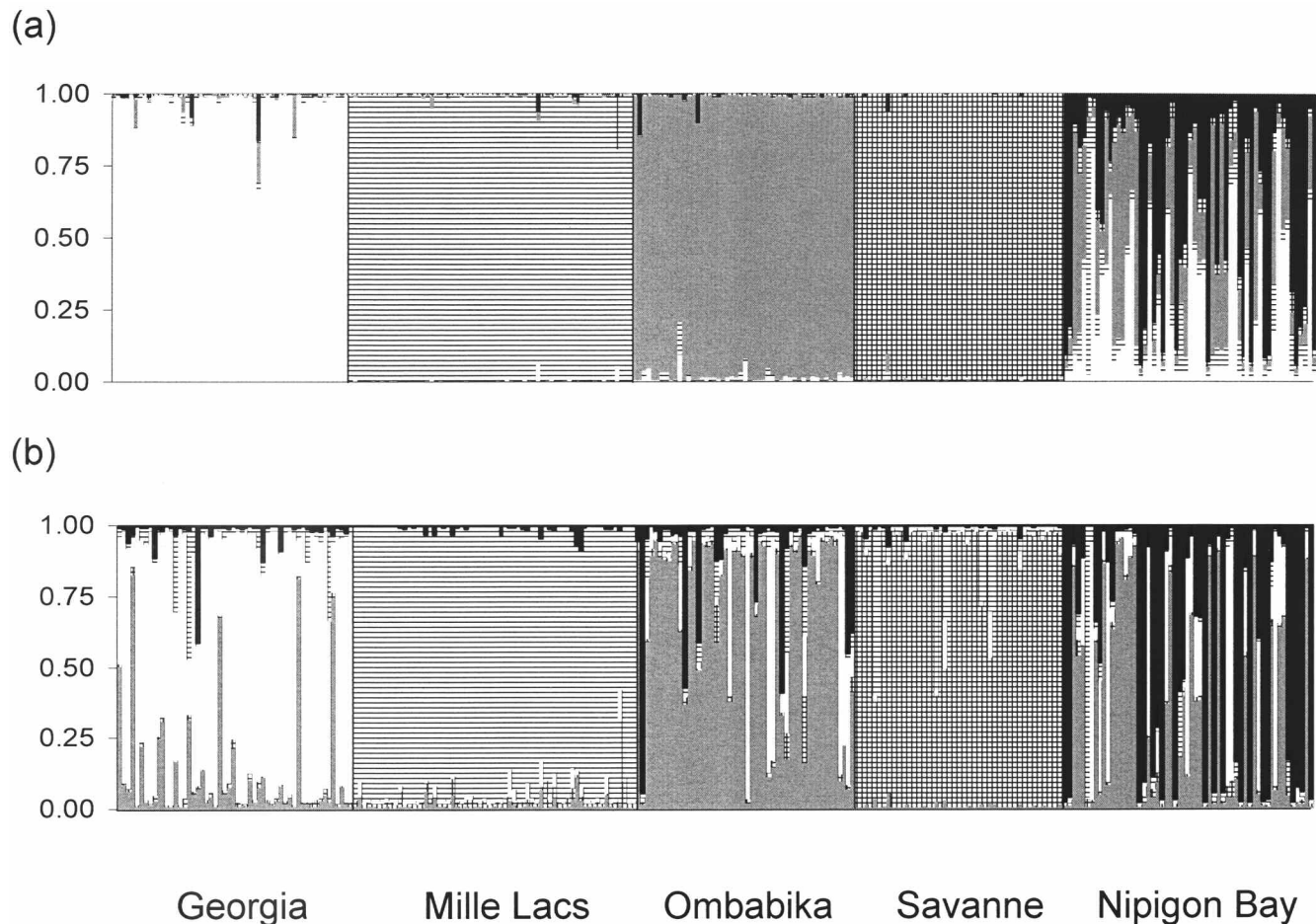


FIG. 4. Genetic ancestry of walleye from source populations and Nipigon Bay, inferred using *STRUCTURE* 2.1 (Pritchard et al. 2000), with (a) and without (b) using a priori population membership information for assignment of individuals from source populations. Vertical bars show fractional membership coefficients for walleye calculated using *STRUCTURE* 2.1 (Pritchard et al. 2000), using the hypothesis of five genetic groups.

ment tests, confirming the close genetic affinity between the historical Black Bay population and contemporary walleye in the Black Sturgeon River (Table 4, Fig. 6). Separating the data into historical, above-barrier and below-barrier groups did not result in significant differences in allelic diversity or frequencies, suggesting that these populations share a common ancestry (Table 4). The above- and below-barrier river samples clustered closely with the historical population, and branch separations among the three sample sets were not significant. By contrast, the Nipigon Bay unknowns were clearly divergent with high bootstrap support (Fig. 6).

DISCUSSION

The data from these studies are encouraging for the rehabilitation prospects of walleye in both Nipigon Bay and Black Bay, and highlight the value of using genetic markers as management and assessment tools for restoration efforts. In both instances, microsatellite DNA markers were highly effective heritable tags for identifying novel (wild) as well as introduced ancestral sources of walleye captured in the two systems. Quantifying the genetic differences among allopatric source populations for adult transfers into Nipigon Bay (Ihssen *et al.* 1993) facilitated subsequent evaluation of establishment success of the introduced strains (Table 3, Fig. 4).

TABLE 4. Allele frequencies and sample sizes for four microsatellite loci for the historical Black Bay population, above- and below-barrier Black Sturgeon River walleye, and unknown walleye from Nipigon Bay.

	Black Bay (historical)	Black Sturgeon R (below dam)	Black Sturgeon R (above dam)	Nipigon unknowns
N	26	26	59	22
Svi2				
192	0.13	0.23	0.14	0.43
194	0.47	0.45	0.41	0.13
196	—	—	0.03	0.03
198	0.03	—	0.01	—
200	0.13	0.18	0.19	—
204	0.23	0.11	0.21	—
206	—	0.02	0.01	0.40
220	—	—	—	0.03
Svi4				
99	—	—	—	0.03
103	—	—	0.01	—
107	0.41	0.32	0.34	0.38
109	0.20	0.09	0.18	—
111	0.15	0.25	0.18	0.05
113	0.22	0.18	0.22	0.55
115	0.02	0.14	0.06	—
117	—	0.02	0.01	—
Svi6				
122	0.26	—	—	—
124	0.52	0.80	0.82	0.55
126	—	—	0.01	—
130	0.04	—	—	—
132	0.10	—	0.02	—
134	0.04	0.14	0.12	0.05
136	0.04	0.07	0.04	0.10
146	—	—	—	0.30
Svi14				
160	—	—	—	0.03
178	—	0.07	0.05	—
182	0.31	0.41	0.42	0.03
184	—	0.14	0.11	0.03
186	—	—	0.01	0.03
188	0.38	0.18	0.19	0.90
190	0.13	0.02	0.08	—
192	—	0.07	0.01	—
194	0.13	0.02	0.06	—
196	0.06	0.02	0.03	—
200	—	0.05	0.03	—
202	—	0.02	0.01	—

For Black Bay, identifying members of the remnant stock as well as possible factors that have impeded the population's recovery are significant steps toward a rehabilitation plan (Krueger *et al.* 1981, Meffe 1995, Colby and Foster 2001).

Applying genetic markers as multigenerational assessment tools similarly highlighted the effectiveness of using an adaptive management framework for restoration and rehabilitation efforts. In cases of population extirpation, the recommended conservation action is to introduce a closely related strain or population, based on the assumption that these will be similarly adapted or suited to local habitats (Krueger *et al.* 1981, Meffe 1995). In some cases, in the absence of pre-existing data, geographic proximity may be used as a surrogate for genetic relatedness, assuming a relationship between geographic and genetic separation. Alternatively, donor sources may be sought that should perform well under the existing ecological and environmental conditions, relying on contemporary adaptations rather than historical relatedness (Moritz 1999, Jones 2003). In the absence of known genetic relatives, it is advisable to introduce several different strains with differing characteristics and follow up with field assessments to evaluate their relative success (Krueger *et al.* 1981, Jones 2003). The substantial differences in establishment and reproductive success among the introduced populations similarly underscored the importance of an adaptive management approach using multiple source populations. Similar efforts for restoring lake trout to several sites in the Great Lakes have shown large differences in survival and establishment among different introduced stocks, and similarly reinforce the effectiveness and value of this approach (Perkins *et al.* 1995, Page *et al.* 2003).

Without genetic information or samples from the historical population, it is not possible to evaluate whether genetic or ecological similarity between the historical and introduced populations were more important. Of the stocking sources used, walleye from Ombabika Bay might be expected to be most ecologically suited to Nipigon Bay due to similarities in habitat such as water body size, temperature, water clarity, and forage base (Wilson 1991). Regardless, the marked differences in performance among the introduced strains underscore the value of testing several candidate strains rather than relying on single-source introductions.

The evidence of a fifth group of walleye in the recovering Nipigon Bay population (Table 3, Fig. 4) is encouraging. The evidence suggests that,

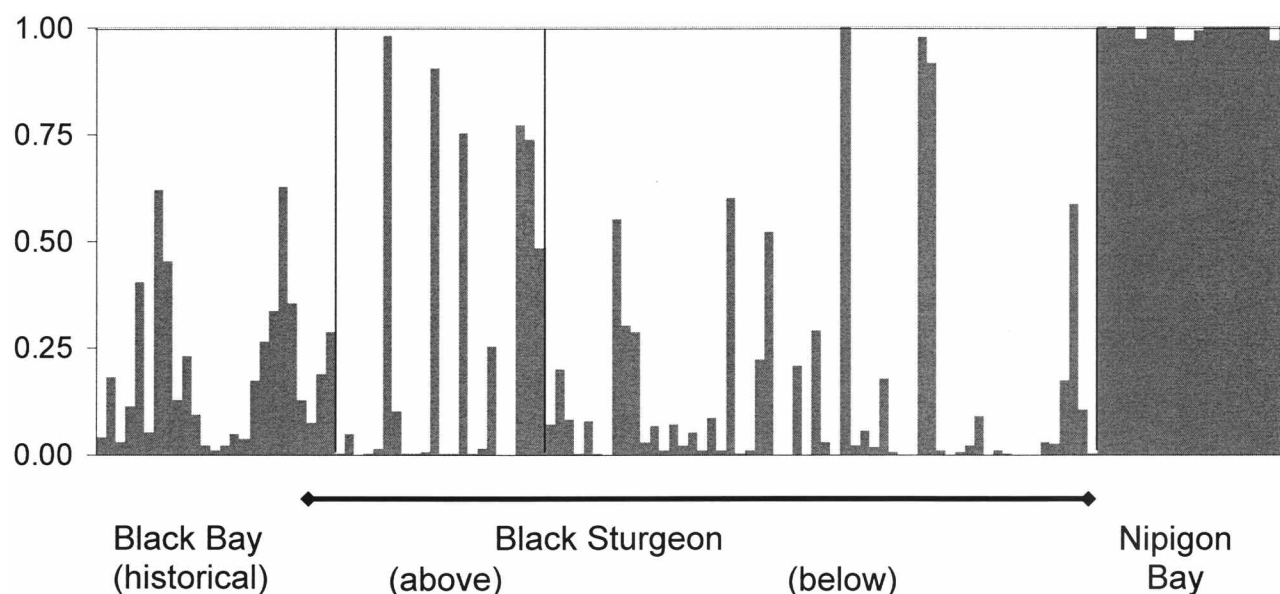


FIG. 5. Fractional membership coefficients for individual walleye from the historical Black Bay population, the Black Sturgeon River (above and below the barrier dam), and Nipigon Bay. Sampling location information was not used to infer a priori population membership. Vertical bars show fractional membership coefficients for walleye calculated using STRUCTURE 2.1 (Pritchard et al. 2000), using the hypothesis of two genetic groups.

given access to suitable restored habitat and release from fishing pressure, walleye are capable of some degree of recovery outside of active intervention efforts such as stocking or adult transfer. Nonetheless, relying on population recovery by habitat mitigation and fisheries closure alone may not result in success, as previous assessment efforts failed to de-

tect significant numbers of walleye despite repeated sampling over multiple years (Wilson 1991).

Whether these unknown walleye from Nipigon Bay represent a surviving remnant of the original population, a successful introduction from other earlier sources, or strays from another wild population in Lake Superior cannot be resolved at present. Although the results from the Nipigon Bay assessment in conjunction with previous field sampling may indicate that adult transfers were more effective than stocking earlier life stages (Wilson 1991), this would not rule out undocumented survival from a previous stocking event. Alternatively, Ryder (1968) mentioned the presence of spawning walleye aggregations in Nipigon Bay tributaries, and suggested that walleye in the Nipigon River, Nipigon Bay, and its tributaries comprised a single stock based on tagging data. Although assessment surveys in tributaries such as the Jackfish River in Nipigon Bay did not detect walleye (Wilson 1991), it may be that finding walleye is highly time-dependent in tributary habitats. Further sampling and analysis of walleye from previously-used stocking sources and Nipigon Bay tributaries may help resolve the ancestry of these mystery fish.

The genetic cohesiveness of walleye from the

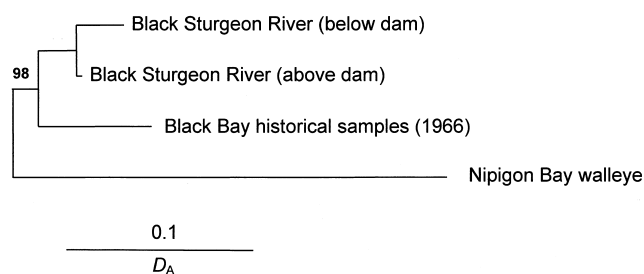


FIG. 6. Neighbor-joining dendrogram of genetic distances among the historical Black Bay population, above- and below-barrier Black Sturgeon River walleye, and unknown walleye from Nipigon Bay based on Nei et al.'s (1983) D_A genetic distance for five microsatellite loci. Numbers on branch nodes represent $\geq 50\%$ percent bootstrap support based on 1,000 resampling events.

historical Black Bay and contemporary Black Sturgeon River populations is remarkable, given the temporal separation of 37 years between sampling periods. The retention of genetic diversity within the Black Sturgeon River watershed nearly four decades after isolation speaks favorably for its holding capacity and production potential.

It is somewhat ironic that in Black Bay, walleye from tributaries were considered to be separate from the exploited population in the bay (R. Ryder, OMNR pers. comm. cited in Schneider and Leach 1977, MacCallum and Selgeby 1987), whereas in neighboring Nipigon Bay the bay and tributary groups were considered to be a single population (Ryder 1968). Although differences in spawning habitat preferences can be heritable in walleye (Jennings *et al.* 1996), the segregation of river- versus shoal-spawning walleye was inferred rather than documented in the Black Bay system (Furlong *et al.* 2006). Similarly, an analysis of river- versus shoal-spawning walleye in the Bay of Quinte, Lake Ontario with microsatellite and mitochondrial DNA markers failed to detect significant differences among sampled spawning aggregations (Wilson and Gatt 2001).

The genetic continuity between the historical lake and contemporary river walleye suggest a similar continuum in habitat use by the former Black Bay population. The Black Sturgeon River watershed likely has sufficient spawning and nursery habitat to produce large numbers of juvenile walleye, but only a finite quantity of space for foraging adults (Furlong *et al.* 2006). Conversely, Black Bay has historically demonstrated its capacity to support a large adult population with abundant forage (MacCallum and Selgeby 1987). A parallel example of interdependency on lake and river habitats occurs in coaster brook trout (*Salvelinus fontinalis*), where tributary rivers provide spawning and nursery habitat and the lake environment provides enhanced opportunities for growth (Schreiner *et al.* 2007, Wilson *et al.* 2007). Re-establishing connectivity among component habitats could greatly enhance walleye numbers in Black Bay and potentially eliminate the need for stocking or transfers. Given the observed proportion of non-introduced walleye in Nipigon Bay with improved habitat conditions and reduced fishing pressure, it may be that walleye in Black Bay will similarly rebound on their own if provided with a restored habitat continuum, low anthropogenic pressures, and time.

Rehabilitation of walleye in northern Lake Superior is still at an early stage, and will require ongoing

efforts for some time to come. Although the early results from Nipigon Bay are encouraging and factors for mitigation have been identified in Black Bay, recovery of both populations is still at least several generations away. Monitoring and assessment programs will be essential elements for monitoring population trajectories and evaluating effectiveness of current and future rehabilitative strategies, and this study demonstrates the value of including genetic tools as part of these efforts. Future rehabilitation efforts should use heritable markers to track strain fitness and success across multiple life stages and generations. Similarly, adaptive management approaches to restoration or rehabilitation may become increasingly important for effectively mitigating anthropogenic or environmental stressors on native species and populations. It is our hope that the evaluative approach used in these studies will be similarly employed in other rehabilitation efforts.

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